

The Architect's Mind in Orbit

Classroom workshop · Orbital Architecture · KTH & ESERO · v2.1.2

For educators running the session: use *teacher-notes-en.pdf* on the day. This sheet is for coordinators, schools, and train-the-trainer briefings.

Summary

A **~2-hour** hands-on workshop (105–130 min as written; ~90–100 min with the optional parts cut) for **ages 12–18** (pilot: Grade 8/9). Students use **their own phones**; the teacher needs a **projector** and **whiteboard**. The theme is how space-station **architecture** affects crew **cognition** — measured with the same kinds of cognitive tests (reaction time, multitasking) that the research team has used at Mars analogue stations and on the ISS.

Live app: magkosm.github.io/orbarch-edu · Languages: EN / SV / EL (?lng=sv / ?lng=el)

Learning objectives

- Name environmental factors that raise or lower stress — in space and in everyday life.
- Explain how scientists measure cognition and compare conditions (baseline vs. stressor).
- Apply design trade-offs: crew cognition vs. mission science and engineering.

Session flow

Block	Time	What happens
A · Presentation	~35–45 min	Station history, ISS reality (Wandt's Destiny-lab photo), whiteboard stress discussion, cognitive link, how we test cognition, MDRS & ISS research context.
B · Experiments	~45–50 min	Reaction Time warm-up → MATB taught on projector → student dry run → Condition Lab (quiet vs. noise) → learning-effect discussion.
C · Design studio	~20–30 min	Architecture Simulator → Habitat Blueprint Designer (trade-offs, corridor-to-entrance rule) → optional Model Lab or A3 sketch.
D · Close	~7 min	Two sticky notes per student (helps / hurts cognition) · closing slide · thanks.

What you need

- Projector + slide deck (paste from *slides-en.md*; structure in *slides-outline.md*)
- Wi-Fi; student phones (Chrome preferred over Safari)
- Earphones for students (the MATB communications task plays spoken radio calls)
- Printed QR sheet — **one QR on screen at a time**
- A3 handouts (1 per student), 2 sticky notes each, whiteboard markers
- Room speakers for cabin noise (or in-app noise in Condition Lab)
- Teacher laptop to mirror a high-scoring Blueprint layout

Print materials (materials/pdf/)

- **workshop-overview-en.pdf** — this sheet
- **teacher-notes-en.pdf** — day-of run sheet
- **handout-en.pdf** — student worksheet (A3)
- **qr-en.pdf** — tool QR codes (A4)

Closing messages

- **Space is for everyone.**
- **Architecture is about people.**
- **Design is a series of trade-offs.**
- **You are the future of space exploration.**

Full plan: [docs/workshop/workshop-en.md](#) · **Pilot feedback:** [docs/workshop/workshop-feedback.md](#)